Course Code: V20CSTPE11 **V20** HTNO | 2 | 2 | A | 8 | 1 | A | 0 | 6 |

SRI VASAVI ENGINEERING COLLEGE(Autonomous)

B.Tech VII Semester Regular and Supplementary Examinations – November-2025

DEEP LEARNING

(Common to CSE & CST)

Time: 3:00 Hrs Max. Marks: 70Marks

Answer All the Questions.
Each Questions Carry Equal Marks

1. **14 M**

A. i. Describe Neural Network in Deep Learning. CO1-K2(7M)
ii. Explain "Complexity and Real-World Impact" on Deep Learning. CO1-K2(7M)

OR

B. i. Explain the historical trends in deep learning. CO1-K2(7M)
ii. Explain challenges arise in training deep neural networks on massive datasets and the applications of deep learning. CO1-K2(7M)

2. **14 M**

A. i. Explain why single-layer perceptron does not solve the XOR problem and how it will be solved with multilayer architectures. CO2-K2(7M)
ii. Differentiate between batch, mini-batch, and stochastic gradient descent? CO2-K2(7M)

OR

B. i. Describe the purpose of hidden units in deep neural network and the identification of the optimal number of hidden units for a task. CO2-K2(7M)
ii. Illustrate the back propagation algorithm and vanishing gradient problem in backpropagation. CO2-K2(7M)

3. **14 M**

A. i. Describe the purpose of regularization in deep learning and the types of regularization. CO3-K2(7M)
ii. Explain the concepts of Early stopping and drop outs in deep neural network. CO3-K2(7M)

OR

B. i. Illustrate the challenges in optimizing deep neural networks. CO3-K2(7M)
ii. Differentiate adaptive learning rate algorithms like Adagrad, Adadelta, and RMSprop with Stochastic Gradient Descent. CO3-K2(7M)

4. **14 M**

A. i. Explain the convolution operation and how does it work in a neural network. CO4-K2(7M)
ii. Explain pooling, its use and various types in the context of CNNs. CO4-K2(7M)

OR

B. i. Explain neuroscientific findings that supports the use of local connectivity in CNN. CO4-K2(7M)
ii. Explain the historical breakthroughs led to the development of Convolutional Neural Networks. CO4-K2(7M)

5. **14 M**

A. i. Explain Encoder-Decoder Sequence-to-Sequence Architectures in detail. CO5-K2(7M)
ii. Explain the concept of Bidirectional RNN ~~with neat diagram~~. CO5-K2(7M)

OR

B. i. Explain LSTM with neat diagram and the mention the disadvantage of RNN. CO5-K2(7M)
ii. Differentiate between GRU and LSTM. CO5-K2(7M)

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